

Detailed GenAI Roadmap (Step-by-Step)

Phase 0: Prerequisites

Core Skills

- Python (functions, classes, virtual environments)
- REST APIs (requests, JSON handling)
- Git (clone, commit, push)
- Basic CLI usage

Key Concepts

- HTTP methods (GET, POST)
- JSON structure
- API keys and authentication

Tasks

- Write a Python script to call a public API
- Parse and display JSON data
- Build a simple CLI tool

Phase 1: LLM Fundamentals

Concepts

- What is an LLM
- Tokens, context window
- Temperature, max tokens
- Prompt structure (system, user, assistant)

Tools

- OpenAI API or equivalent

Tasks

- Generate text using API
- Build Q&A system
- Implement summarization
- Experiment with prompt variations

Mini Project

- CLI-based chatbot

Phase 2: LangChain Basics

Concepts

- Prompt templates
- Chains
- Output parsers

Tasks

- Create reusable prompt templates
- Build sequential chains
- Format structured outputs (JSON)

Mini Project

- Input → Summarize → Translate → Format pipeline

Phase 3: Retrieval-Augmented Generation

Concepts

- Embeddings
- Vector databases
- Chunking strategies
- Similarity search

Tools

- FAISS / Pinecone / OpenSearch

Tasks

- Convert text into embeddings
- Store and retrieve vectors
- Implement retrieval pipeline

Mini Project

- Chat with PDF system:
 - Upload document
 - Ask questions
 - Retrieve relevant context
 - Generate grounded answers

Phase 4: LangChain Agents

Concepts

- Agents and reasoning
- ReAct pattern (Thought → Action → Observation)

Tasks

- Create basic agent
- Integrate tools (calculator, search)

Mini Project

- Assistant that answers queries

Phase 5: Agents with Tools

Concepts

- Custom tool creation
- Function calling
- Tool selection logic

Tasks

- Build custom tools (API caller- Weather API, News API, Search API, Calculator)
- Connect multiple tools to the agent
- Improve tool accuracy

Mini Project

- Agent that:
 - Answers questions
 - Uses a calculator when needed

Phase 6: Agentic Workflows

Concepts

- Task decomposition
- Planning and execution
- Memory handling
- Iterative refinement

Tasks

- Build multi-step workflows
- Implement memory (short-term context)
- Add validation/reflection steps

Mini Project

- Research assistant:
 - Breaks question
 - Searches data
 - Summarizes results
 - Refines output

Phase 7: LangGraph

Concepts

- Graph-based workflows
- Nodes and edges
- State management
- Conditional routing

Tasks

- Define workflow graphs
- Implement branching logic
- Maintain state across steps

Mini Project

- Stateful chatbot with:
 - Multi-step reasoning
 - Decision-based flows

Phase 8: Model Context Protocol

Concepts

- Tool servers
- Structured tool exposure
- Standardized tool calling

Tasks

- Build MCP server
- Define tools (APIs, functions)
- Connect LLM agent to MCP tools

Mini Project

- MCP-based system:
 - Tool server with multiple tools
 - Agent consuming tools dynamically